

NuTureSense

Research Questions

Project focus: Secure, adaptive communication for low-power sensing using RuView / Wi-Fi CSI, LoRa, and authenticated encryption.

1. Central Research Question

CRQ: How can RuView / Wi-Fi CSI sensing data be securely transmitted over a low-power, long-range LoRa link using an adaptive authenticated-encryption framework while keeping security and communication overhead practical?

2. Primary Research Questions

RQ1. Can RuView / Wi-Fi CSI sensing data be securely transmitted over a LoRa communication link using authenticated encryption?

RQ2. How do AES-GCM and ChaCha20-Poly1305 differ in encryption/decryption performance and resource overhead on the target platform?

RQ3. Can the NuTureSense security layer detect and reject packet tampering, replay attacks, and forged packets?

RQ4. Can an adaptive cryptographic controller safely select between AES-GCM and ChaCha20-Poly1305 without breaking authentication, nonce management, key management, or replay protection?

RQ5. What performance overhead does the security layer introduce compared with an unsecured LoRa communication system?

3. Security Research Questions

SQ1. How effectively does authenticated encryption prevent unauthorized modification of sensing packets?

SQ2. How does the framework detect and reject replayed packets using sequence or timestamp information?

SQ3. Can an attacker without the session key construct a packet that passes authentication?

SQ4. How are nonces generated and managed so that each AEAD packet receives a unique nonce?

SQ5. How does the protocol maintain authentication and replay protection when the selected cipher changes?

SQ6. What security assumptions, trust boundaries, and out-of-scope threats should be defined for the NuTureSense threat model?

4. Adaptive Cryptography Research Questions

AQ1. What criteria should the security controller use to select AES-GCM or ChaCha20-Poly1305?

AQ2. How do available computation, energy budget, hardware acceleration, and configured security policy affect cipher selection?

AQ3. What protocol information must be authenticated so that both sender and receiver remain synchronised after a cipher switch?

AQ4. What happens if a packet containing a cryptographic state transition is lost or delayed?

AQ5. Does adaptive selection improve resource or communication efficiency compared with using a single fixed AEAD cipher?

5. Performance and Communication Research Questions

PQ1. What is the encryption and decryption latency for AES-GCM and ChaCha20-Poly1305?

PQ2. How much additional packet overhead is introduced by security metadata, nonces, sequence information, and authentication tags?

PQ3. How does authenticated encryption affect end-to-end throughput?

PQ4. How does the security layer affect packet delivery, packet loss, and communication reliability over LoRa?

PQ5. What are the CPU, memory, and energy costs of the two AEAD algorithms on the target hardware, where these measurements are available?

PQ6. How does the security configuration affect communication performance under different LoRa conditions?

6. Experimental / Evaluation Questions

EQ1. What is the baseline performance of the system without encryption?

EQ2. How does the baseline compare with AES-GCM, ChaCha20-Poly1305, and the adaptive configuration?

EQ3. What is the detection rate for tampering, replay, and forgery test cases?

EQ4. What security and performance trade-offs are observed across the tested configurations?

EQ5. Are the measured results consistent across repeated experiments and different test conditions?

7. Research Question → Measurement Mapping

Question Area	Main Measurements / Evidence
Secure transmission	Authentication success/failure, confidentiality test, packet validation
Tampering	Modified-packet rejection rate
Replay	Replayed-packet detection/rejection rate
Forgery	Invalid-authentication rejection rate
AES-GCM vs ChaCha20-Poly1305	Encryption time, decryption time, CPU, memory, energy where measurable

Question Area	Main Measurements / Evidence
Communication impact	Latency, throughput, packet size, packet delivery/loss, RSSI/SNR where available
Adaptive mechanism	Correct cipher selection, synchronisation, switch handling, security after switching

8. Recommended Core Set for the Project Report

For the main research paper, use CRQ plus RQ1–RQ5 as the core research questions. Use the Security, Adaptive Cryptography, Performance, and Evaluation questions as supporting sub-questions that guide implementation and experiments.

Source basis: The questions are derived from the supplied NuTureSense framework, which defines the sensing, security, authenticated-encryption, LoRa communication, receiver validation, attack testing, and evaluation components.